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[27]

SARDAR PATEL UNIVERSITY
F.Y B.C.A (2nd Sem) EXAMINATION (NC) APRIL-2026
MATHEMATICS || US02SBCA51

Date: 16/04/2026

Time: 10:00 AM to 12:00 PM

Total Marks:50

Q-1 Select Appropriate Answer.

[10]

- 1 Dot product of $u = (1, 2, 3)$, $v = (0, 1, 4)$ is ____
(a) 14 (b) 2 (c) $(1, 1, 7)$ (d) 10
- 2 A Square matrix A is said to be symmetric if ____
(a) $A \neq A^T$ (b) $A = -A^T$ (c) $A = A^T$ (d) None
- 3 An Edge whose endpoints are the same vertex is called ____
(a) trivial (b) multigraph (c) loop (d) multiple
- 4 The matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is ____
(a) Identity (b) Row (c) Column (d) Zero
- 5 A vertex does not belongs to any edge is called
(a) Trivial (b) Isolated vertex (c) Nullgraph (d) Finite
- 6 A graph G is ____ if each vertex is connected to every other vertex.
(a) Disconnected (b) Closed (c) Complete (d) Connected
- 7 A map is a particular _____ representation of a finite planar multigraph.
(a) planar (b) non planar (c) complete (d) simple
- 8 The median of 2, 4, 6, 1, 5, 3, 7 is ____
(a) 3 (b) 4 (c) 5 (d) None
- 9 The value which occurs with the maximum frequency is called ____
(a) Median (b) Mode (c) Mean (d) None
- 10 _____ is the value dividing the observations in two equal parts.
(a) Mode (b) Median (c) Mean (d) None

Q-2 Short Questions (Attempt 8 out of 10)

[16]

- 1 If $u = (1, 4, 3, 9, -3)$, $v = (-5, -2, 5, 6, 8)$, then evaluate $\|u\|$ and $\|v\|$
- 2 Find x and y if, $x(1, 1) + y(2, -1) = (1, 4)$

- 3 Find the value of k such that vectors $(5, 4, k)$ and $(3, -4, 1)$ are orthogonal.
- 4 If $u = (1, 4, 3)$, $v = (-5, -2, 5)$, then evaluate (1) $u+2v$ (2) $u \cdot v$
- 5 Draw a diagram of the complete graphs K_5 , K_6 .
- 6 Draw a picture of each of the following graphs, and state whether or not it is simple.
 $G = (V_2, E_2)$, where $V_2 = \{P, Q, R, S, T\}$ and $E_2 = \{PQ, PR, PS, PT, TR, PR\}$.
- 7 Draw a picture of each of the following graphs, and state whether or not it is simple.
 $G = (V_1, E_1)$, where $V_1 = \{a, b, c, d, e\}$ and $E_1 = \{ab, bc, ac, ad, de\}$.
- 8 Explain the terms: Mode.
- 9 Find the Mean of 2, 4, 8, 12, 16 and 24.
- 10 Find Mean of the following data:

X	0	1	2	3	4	5	6
F	2	8	20	4	5	10	30

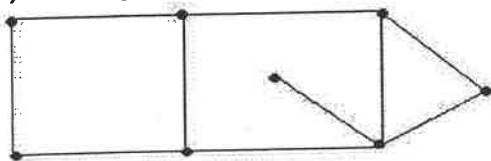
Q-3 If $A = \begin{bmatrix} 1 & 3 \\ 7 & 0 \end{bmatrix}$ evaluate $f(A)$ for the polynomial where $f(x) = 2x^2 + 4x + 5$ [06]

OR

Q-3 For given matrices find AB and BA [06]

$$\begin{bmatrix} 2 & 5 \\ 5 & 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

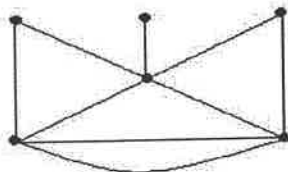
Q-4 (a) Find cycle or closed path for given graph. [06]



(b) Find the number m of edges in the graphs: 1) K_{120} 2) K_{200}

OR

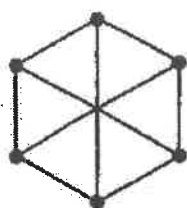
Q-4 For the following graph: [06]



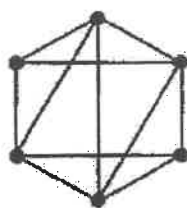
- (a) Describe graph formally.
- (b) Find degree of all vertices and parity of each vertex.
- (c) Is it true "The sum of each vertices is equal to twice the number of edges."

[06]

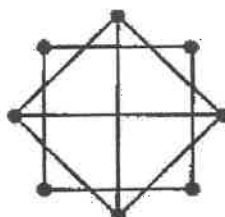
Q-5 Define: Planar graph. Checks which of the following are planar graphs. Justify.



(a)



(b)



(c)

OR

Q-5 Calculate Mean and Median for the following data.

[06]

Age	0-10	10-20	20-30	30-40	40-50
Frequency	4	6	12	8	5

[06]

Find the Median & Mode of the given data:

Q-6

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No. of Students	2	5	8	16	9	5

OR

Q-6 The mean of 200 items was 50. Later on it was found that two items were wrongly read as 92 and 8 instead of 192 and 88. Find out the correct mean.

[06]

— × —

(3)

